

# Modifier modding

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**Modifier modding** involves techniques that allow for application of modifier types to countries and other entities. These modifiers then help to define what the nation can do and with what efficiency.

Please help with verifying or updating older sections of this article.  
At least some were last verified for version 1.0.





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# Static modifiers

Static modifiers are the backbone mechanism for adding dynamic modifiers during gameplay. They are easily added and removed via [effects](#) and provide utility in terms of AI and UI support.

## Technical details

Static modifiers are located in `common/static_modifiers`, usually in the `main_menu` [top folder](#).

For example:

```
common/static_modifiers/example_file.txt
```

## Making modifiers

A basic static modifier is comprised of a set of modifiers and the values assigned to them:



```
modifier = {  
    country_cabinet_efficiency = 0.1
```



The values on the right side should match what the modifier type on the left expects, a number or a boolean. The value can also be a static scripted value; that is, it must evaluate to a number or boolean directly – it cannot be dynamically calculated. For example:

```
my_modifier = {
    modifier_type_1 = test_value_1    # 0.2 underneath
    modifier_type_2 = test_value_2    # will NOT work, a dynamic value is underneath.
}
```

With the following script value definitions:

```
test_value_1 = 0.2                # will work
test_value_2 = {                  # will NOT work - no calculations on the fly
    value = 0.1
    multiply = war_exhaustion
}
```

Besides modifier types, a static modifier also accepts a `game_data` field.

## Game data

Game data contains two fields; the required `category` and the optional `decaying`.

`category` is used to determine the scope of the modifier. If it is set to `country`, then the modifier is applied on countries using `add_country_modifier`. Here are the following available categories:

- `character`
- `country`
- `international_organization`
- `location`
- `mercenary`
- `province`
- `rebel`
- `religion`
- `unit`

`decaying` is an optional attribute which, when set to `yes`, reduces the effects of the modifiers based on remaining time in a linear fashion. If a modifier is 50% towards its expiration, it has 50% of the strength.

Example:

```
test_modifier = {
    game_data = {
        decaying = yes
        category = country
    }
}
```



## Applying and removing modifiers during the game

Modifiers can be applied during the course of the game by using `add_<type>_modifier = {` effects, which all follow the same general syntax:

```
add_<type>_modifier = {
    mode = add_and_replace
    modifier = <key>
    years = 25
    # and others... See Add modifier parameters
}
```

Modifiers can be removed using `remove_<type>_modifier` effects, for instance `remove_country_modifier`.

Here is the full list of modifier effects:

List of add modifier effects [\[Genişlet\]](#) List of remove modifier effects [\[Genişlet\]](#)

### Add modifier parameters

The most important aspect of any modifier addition is the `modifier` which determines which modifier gets added. Modifier addition also needs to have a duration set by either `years`, `months`, `weeks` or `days`, which all accept scripted values evaluated based on the current scope and saved scopes. If the duration evaluates to `-1`, the modifier will be permanent.

The `mode` is an important part of defining how the modifier addition will function, and accepts the following:

| Mode type                        | Behavior                                                                                                                                             |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>replace</b>                   | Removes the modifier and reapplies it for the whole duration provided.                                                                               |
| <b>add</b>                       | Duration will not be changed and the modifiers will be added onto existing modifier.                                                                 |
| <b>extend</b>                    | Duration of an existing modifier will be extended by the provided duration.                                                                          |
| <b>add_and_extend</b>            | Stacks the modifiers together and extends the length by the duration.                                                                                |
| <b>set_to_largest</b>            | If the modifier strength ( <code>size</code> ) is stronger than the current modifier applied, set it to new size.                                    |
| <b>set_to_largest_and_extend</b> | If the modifier strength ( <code>size</code> ) is stronger than the current modifier applied, set it to new size. Additionally, extend the duration. |

The default is `extend`.

The strength of the modifier added can also be dynamically calculated using the `size` parameter, which accepts a script value, calculated similarly to how the duration value is calculated.

If one wishes to override the text that is shown in the modifier addition text, one may use `desc` to set a localizable description:



```

desc = head_of_cabinet_influence_desc
}

```

This results in text like "We gain <modifier> for <head\_of\_cabinet\_influence\_desc>"

Lastly, `recalculate_immediately = yes` can be used to force the game to recalculate all modifiers after the effect. This will make the game recognize new changes immediately.

## Applying modifiers in startup

*Main article: [Setup modding#Timed modifiers](#)*

Static modifiers can also be applied to various game objects during startup using `timed_modifiers`.

## Localisation

*Main article: [Localization](#)*

To localize a static modifier, `STATIC_MODIFIER_NAME_<key>` must be localized. `STATIC_MODIFIER_DESC_<key>` is also supported to represent the description, but can be left empty (set as "").

Defined modifiers can be retrieved for localization and GUI using `ShowModifier('modifier_name')` and `ShowModifierWithNoTooltip('modifier_name')`.

## Triggers

There are two types of triggers related to modifiers: one that allows for checking their presence, for instance `has_country_modifier` and its equivalents for other types.

The other, `add_static_modifier_utility` is a complex trigger that returns a numerical value representing the AI's perceived value of the adding the static modifier provided. It accepts country, character and character scopes:

```
value = "add_static_modifier_utility(joined_order_of_the_band)"
```

Additionally, a second numerical argument can be provided to indicate the size parameter of the modifier to be evaluated:

```
value = "add_static_modifier_utility(joined_order_of_the_band|0.5)"
```

`remove_static_modifier_utility` is opposite version of the above, utilizing the same syntax but calculating the value of removing the static modifier instead.

List of modifier triggers [\[Genişlet\]](#)

## dcoded static modifiers



[Genişlet]

## Auto modifiers

Auto modifiers are automatically applied to countries or international organizations when certain criteria are met and can be scaled at will. They need not be explicitly added to every country, as they are automatically applied if possible.

Because of this, they need to be used sparingly as they can have serious performance consequences if misused.

### Technical details

Auto modifiers are located in `common/auto_modifiers`, usually in the `in_game` top folder.

### Auto modifier example

Here is an example of an auto modifier:

```
war_exhaustion_impact = {
    scales_with = war_exhaustion

    land_morale_modifier = -0.02
    naval_morale_modifier = -0.02

    global_production_efficiency = scaled_production_efficiency_penalty
    trade_efficiency = scaled_trade_efficiency_penalty
    global_population_growth = -0.0003
}
```

### Auto modifier category and scope

By default, all auto modifiers are country modifiers based on country scope. The modifier category and scope type can be adjusted using `category` and `type` respectively:

```
auto_modifier_example = {
    type = international_organization
    category = international_organization
    ...
}
```

Currently, the only other supported type is `international_organization`.

### Additional parameters

`requires_real = no` is used to apply the country auto modifier on "non-real" countries like the Pirate, Rebel or Mercenary tags.



triggers and scaling

To check whether an auto\_modifier should appear, use potential\_trigger and limit. Both conditions must evaluate to true for the auto\_modifier to apply; it will be removed when either becomes false.

Based on empirical testing, the auto\_modifier limit field only supports basic inequality checks (<, >, <=, >=) and is likely more efficient for evaluating these conditions. Some examples of what do not work within the limit field are =, AND, OR, and NOT. Thus, limit = { legitimacy >= 50 } will work correctly, while limit = { NOT = { legitimacy > 50 } } will not, with the result always defaulting to true. The potential\_trigger does not have this limitation and can accept all standard triggers and comparisons.

The strength of the auto modifier is calculated dynamically using scales\_with script value.

Localisation and datacontext

Auto modifiers can be localized similarly to static modifiers under the following strings:

- AUTO\_MODIFIER\_NAME\_<key> for the title
- AUTO\_MODIFIER\_DESC\_<key> for the desc

Auto modifiers can also be used in datacontext with the following:

- StaticAutoModifier Type
- ShowAutoModifierEffect( 'auto\_modifier' ) Function
- ShowAutoModifierEffectForCountry( 'auto\_modifier' ) Function
- ShowAutoModifierEffectForLocation( 'auto\_modifier' ) Function

New modifier types

Main article: [Modifier types](#)

Unlike in Europa Universalis IV modding, new modifier types can be added in common/modifier\_type\_definitions. They are best placed in the main\_menu top folder.

Their values can then be used in multiple ways, but are most commonly referenced using modifier: [datalink](#).

References

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